

Task 3: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
saikrishna@DESKTOP-LGFVDTJ:~$ nano practicalweek3.c
saikrishna@DESKTOP-LGFVDTJ:~$ gcc practical.c -o practicalweek3
cc1: fatal error: practical.c: No such file or directory
compilation terminated.
saikrishna@DESKTOP-LGFVDTJ:~$ gcc practical.c -o practicalweek3
cc1: fatal error: practical.c: No such file or directory
compilation terminated.
saikrishna@DESKTOP-LGFVDTJ:~$ gcc practicalweek3.c -o practicalweek3
saikrishna@DESKTOP-LGFVDTJ:~$ ./practicalweek3
Parent process started
Parent PID: 570
Parent PPID: 326

Parent Process
Parent PID: 570
Child PID: 571
Parent State: RUNNING

Child Process
Child PID: 571
Child PPID: 570
Child State: RUNNING
Child State: TERMINATED
Parent State: RUNNING after child completion
saikrishna@DESKTOP-LGFVDTJ:~$ cat practicalweek3.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID: %d\n", getpid());
    printf("Parent PPID: %d\n", getppid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("Child State: RUNNING\n");

        sleep(3);

        printf("Child State: TERMINATED\n");
    }
}
```

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    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    printf("Parent State: RUNNING\n");

    wait(NULL);

    printf("Parent State: RUNNING after child completion\n");
}

return 0;
}
saikrishna@DESKTOP-LGFVDTJ:~$ ./practicalweek3
Parent process started
Parent PID: 595
Parent PPID: 326

Parent Process
Parent PID: 595

Child PID: 596
Child Process
Parent State: RUNNING
Child PID: 596
Child PPID: 595
Child State: RUNNING
Child State: TERMINATED
Parent State: RUNNING after child completion
saikrishna@DESKTOP-LGFVDTJ:~$ |

```

Task 4: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```

saikrishna@DESKTOP-LGFVDTJ:~$ nano PracticalWeek3.c
saikrishna@DESKTOP-LGFVDTJ:~$ sleep 30 &
[1] 651
saikrishna@DESKTOP-LGFVDTJ:~$ ps
  PID TTY          TIME CMD
  329 pts/0    00:00:00 bash
  651 pts/0    00:00:00 sleep
  661 pts/0    00:00:00 ps
saikrishna@DESKTOP-LGFVDTJ:~$ ps -o pid,ppid,state,cmd
  PID   PPID  S  CMD
  329     324  S  -bash
  651     329  S  sleep 30
  673     329  R  ps -o pid,ppid,state,cmd
saikrishna@DESKTOP-LGFVDTJ:~$ top
top - 05:19:47 up 22 min,  1 user,  load average: 0.02, 0.02, 0.00
Tasks: 24 total,   1 running, 23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,
MiB Mem :  7821.6 total,  7331.8 free,   496.4 used,   143.7 buff/c
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  7325.3 avail

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+
708 saikris+  20   0   8160  6004  3896 R   0.3   0.1   0:00.67
  1 root      20   0  24088 15432 11640 S   0.0   0.2   0:01.32
  2 root      20   0   3180  2204  2072 S   0.0   0.0   0:00.01
  7 root      20   0   3180  2048  1940 S   0.0   0.0   0:00.00
 50 root      19  -1 50400 16776 15492 S   0.0   0.2   0:00.33
 85 systemd+ 20   0  22416 14460 11968 S   0.0   0.2   0:00.16
 93 root      20   0  35328 12308  9256 S   0.0   0.2   0:01.09
114 root      20   0   2888  1948  1820 S   0.0   0.0   0:00.07
115 root      20   0   4472  3136  2884 S   0.0   0.0   0:00.02
116 message+ 20   0   8820  5496  4828 S   0.0   0.1   0:00.16
155 root      20   0  44096 29856 14720 S   0.0   0.4   0:00.46
159 root      20   0  18448  9460  8204 S   0.0   0.1   0:00.14
183 syslog    20   0  220548 5816  4516 S   0.0   0.1   0:00.16
196 root      20   0   5492  2720  2536 S   0.0   0.0   0:00.02
212 root      20   0 123460 32620 18008 S   0.0   0.4   0:00.44
219 _chrony     20   0  23700 11132  7092 S   0.0   0.1   0:00.19
240 _chrony     20   0  12096  2404  1760 S   0.0   0.0   0:00.00
322 root      20   0   3188  1108   980 S   0.0   0.0   0:00.00
324 root      20   0   3204  1248  1108 S   0.0   0.0   0:00.11
329 saikris+  20   0   6268  5596  3928 S   0.0   0.1   0:00.09
332 root      20   0   8648  5540  4696 S   0.0   0.1   0:00.02
387 saikris+  20   0  22340 13516 10896 S   0.0   0.2   0:00.30
389 saikris+  20   0  22912  4116  2116 S   0.0   0.1   0:00.00
415 saikris+  20   0   6136  5396  3848 S   0.0   0.1   0:00.11

[1]+  Done                  sleep 30
saikrishna@DESKTOP-LGFVDTJ:~$ q
Command 'q' not found, but can be installed with:
sudo snap install q # version 1.6.3-1, or
sudo apt install python3-q-text-as-data # version 3.1.6-4build1
See 'snap info q' for additional versions.
saikrishna@DESKTOP-LGFVDTJ:~$ cat /proc/1234/status
cat: /proc/1234/status: No such file or directory

```

```
Pid: 415
PPid: 332
TracerPid: 0
Uid: 1000    1000    1000    1000
Gid: 1000    1000    1000    1000
FDSize: 256
Groups: 4 24 27 30 46 100 1000
NSTgid: 415
NSpid: 415
NSpgid: 415
NSSid: 415
Kthread: 0
VmPeak: 6136 kB
VmSize: 6136 kB
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 5396 kB
VmRSS: 5396 kB
RssAnon: 1548 kB
RssFile: 3848 kB
RssShmem: 0 kB
VmData: 1496 kB
VmStk: 132 kB
VmExe: 1036 kB
VmLib: 1908 kB
VmPTE: 48 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffff
Threads: 1
SigQ: 1/31243
SigPnd: 0000000000000000
ShdPnd: 0000000000000000
SigBlk: 0000000000000000
SigIgn: ffffffff3478c004
SigCgt: 000000004b813efb
CapInh: 0000000000000000
CapPrm: 0000000000000000
CapEff: 0000000000000000
CapBnd: 000001ffffffffffff
CapAmb: 0000000000000000
NoNewPrivs: 0
Seccomp: 0
Seccomp_filters: 0
Speculation_Store_Bypass: thread vulnerable
SpeculationIndirectBranch: conditional enabled
Cpus_allowed: ff
Cpus_allowed_list: 0-7
Mems_allowed: 00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000
Mems_allowed_list: 0
voluntary_ctxt_switches: 43
nonvoluntary_ctxt_switches: 1
saikrishna@DESKTOP-LGFVDTJ:~$
```